

CALIBRATE BEFORE YOU RANK: CORRUPTION-VALIDATED QUALITY METRICS FOR RHYTHM-GAME CHARTS

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ABSTRACT

Rhythm-game charts are designed artifacts with many valid realizations per song, yet generated charts are still evaluated with metrics borrowed from other tasks: F-measure against one reference chart, language-model (LM) perplexity, and unreferenced statistics such as entropy, distinct- n , and self-similarity. We validate metrics the way one would test a smoke detector: by setting controlled fires. Applying nine targeted corruption probes at increasing doses to 170 human charts (4,760 scored variants), we find failures invisible to spot checks: perplexity *improves* by 37% when charts are flattened toward the LM’s modal patterns, self-similarity structureness *rises* by 62% under copy-paste collapse, and all 56 chart-only quantities we compute are blind to a global 60 ms audio misalignment that would be fatal in play (max response 0.03 SD). A two-layer protocol repairs these failures: raw diagnostics are normalized against per-difficulty target bands calibrated on the official-chart corpus, and a perceptually anchored timing family—a 6 ms just-noticeable-difference deadzone, severity tiers at its multiples, and a fixed-lattice phase witness over a two-tier grid-reference hierarchy—recovers injected 15/30/60 ms shifts with exact medians on $\geq 99.4\%$ of charts, with no reference chart. The probes also expose complementarity: a sparse-outlier corruption that grid matching largely absorbs is caught at the token level by the grammar family, while the anchor shift invisible to every chart-only metric is caught only by the timing witness—the score profile is not compressible to a scalar. Applied to five published generators across three chart vocabularies, the profile turns single-number rankings into mechanism diagnoses. We release the probes, calibration artifacts, per-chart records, and an installable evaluation package. [PENDING: P8: artifact URL.]

1. INTRODUCTION

Automatic chart generation for rhythm games—mapping audio to a playable sequence of timed, typed notes—has progressed from onset-detection pipelines to conditional

sequence models [1–4]. Evaluation has not kept pace. Because a chart is a creative artifact rather than an annotation, the same song admits many valid charts, and the field’s dominant tool—F-measure against the single official chart—penalizes every valid alternative equally and cannot be computed at all for songs without charts. The natural complement is reference-free measurement, for which authors borrow from symbolic music and text generation: LM perplexity [1], pattern-space congruence [2], event entropy, distinct- n [5], Self-BLEU [6], feature-distribution comparison [7,8], and self-similarity structureness [9].

This paper asks whether those borrowed metrics measure chart quality at all, and answers with a validation method that requires no listening test: *targeted corruption*. Starting from human (official) charts, we apply nine damage operators—dense timing jitter, sparse severe outliers, global anchor shift, color shuffling, copy-paste collapse, LM-argmax blandification, density scaling, burst insertion, bar shuffling—at three doses each. On average, corrupting a human chart cannot make it better, so a usable quality metric must respond in the right direction, and the response must grow with dose. Metric validation thus becomes a fully automated, reproducible procedure that requires neither human raters nor model training, and applies unchanged to any proposed chart metric.

The audit yields a negative result with a constructive core. Borrowed metrics fail structurally (Section 5.2): perplexity rewards the blandness it should punish, structureness mistakes degenerate repetition for form, several statistics are near-proxies for chart length, and every chart-only quantity is blind to global audio misalignment. But the underlying raw quantities carry signal; what is missing is calibration and grounding. We show that (i) normalizing each diagnostic against per-difficulty target bands estimated from the official corpus restores a consistent, gaming-resistant reading, and (ii) a timing family anchored on a grid-reference hierarchy—authored per-bar timing metadata when available, an audio-estimated beat grid otherwise, never a generated chart’s self-declared grid—closes the misalignment blind spot with no reference chart.

Our contributions are:

1. **A corruption-probe audit harness:** nine dose-controlled damage operators with acceptance criteria (population-level monotone response, anti-gaming, published coupling structure) and fairness



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rules, applicable unchanged to any future chart metric (Section 4).

2. **A structural failure analysis** of borrowed metrics: two reductio exhibits (perplexity improves under blandification; structureness improves under collapse), an all-blind result under anchor shift, and a direction-instability result—perplexity ranks correctly against noisier-than-corpus systems yet inverts against blander-than-corpus ones (Section 5).
3. **A two-layer, reference-free protocol** with an empirically assigned role taxonomy and a perceptually anchored timing family: a single 6 ms JND constant grounds a deadzone, severity tiers, and a clean-note rate; a piecewise fixed-lattice witness recovers injected shifts exactly where matching statistics stay blind (Section 3, Section 5.3).
4. **Complementarity evidence**: three corruptions, each invisible to one metric family and glaring to another, in both directions—the empirical case that chart quality is a profile, not a scalar (Section 5.4).
5. **Application to five published systems** across three chart vocabularies, yielding mechanism diagnoses and an applicability matrix instead of a single ranking (Section 5.5), plus released artifacts: an installable package, probe implementations, calibration bands, per-chart records, and grouped song-level dataset splits. **[PENDING: P8: public repo URL.]**

The probes establish what metrics detect: *damage*. Whether the surviving metrics also track player experience is a distinct question that requires perception studies; we return to it in Section 7.

2. RELATED WORK

Chart generation and its evaluation. Dance Dance Revolution evaluates step placement by F-measure against a reference chart and step selection by LM perplexity [1]; later systems inherit this template [3, 4]. TaikoNation adds pattern-congruence measures against human charts [2]. None of these metrics has been audited against controlled damage.

Generative-music evaluation. Yang and Lerch compare feature distributions between generated and reference sets [7]; MusPy popularized symbolic statistics [8]; the Jazz Transformer introduced self-similarity structureness indicators [9]. These are set-level or direction-ambiguous quantities; we audit their behavior when used as per-chart quality signals.

Metric critiques. Theis et al. showed that likelihood and sample quality can diverge arbitrarily [10]; our blandification result is a sharp corpus-LM instance. NLG research documented pathologies of distinct- n and Self-BLEU [5, 6]. Unlike pure critiques we supply the repair, and unlike human-anchored metric studies our harness needs no raters. Onset F-measure follows

`mir_eval` [11]. Timing thresholds follow the psychophysics of temporal discrimination [12, 13] rather than any game’s scoring windows.

3. TWO-LAYER EVALUATION PROTOCOL

Data. We use a corpus of 1,155 taiko-style chart files covering 1,013 unique songs, grouped by audio identity and normalized title and split at the song level into 813/80/120 training/validation/test song groups, so that no song appears in more than one split. Calibration uses the official difficulty-course charts of the training songs; evaluation uses 40 held-out test songs (170 official course-charts) across all difficulty courses.

Layer 1: raw diagnostics. Each chart is a sequence of timed, typed events. We compute quantities with native units and no claimed direction: note density and local-density percentiles, inter-onset entropy, color-switch rate, n -gram LM negative log-likelihood (trigram, add- α smoothing, trained on the calibration corpus), unseen-transition rate, unique/repeated 4-gram rates, and windowed LM surprisal percentiles. These deliberately coincide with the borrowed metrics: the audit compares *normalization strategies*, not measurement choices.

Layer 2: calibrated band scores. For each diagnostic and difficulty course we estimate a target band $[p_{10}, p_{90}]$ over the calibration corpus. Scores are 1 inside the band and decay smoothly outside; error-type quantities use exponential penalties. Band scores measure *typicality*: “more entropy” is neither good nor bad, but “entropy within the range human charters use at this difficulty” is well defined, and the score is reference-free at the song level—scoring a chart for a novel song requires only the fixed corpus calibration.

Two metric types. The suite separates *defect-type* quantities, whose direction is unconditional (transitions that never occur in the human corpus; violations of playability constraints), from *typicality-type* quantities, where only the probes—not the sign of the deviation—justify an interpretation. For typicality scores the inference is one-directional: the probes certify that damage drives charts out of band; they do not certify the converse. An out-of-band reading is a flag whose meaning is adjudicated by the rest of the profile (Section 6).

Role taxonomy. Scores are assigned roles empirically by the audit, not a priori: *rankers* respond to damage and separate systems (pattern-IC, transition validity, repetition and variety adequacy, rhythm complexity); *constraints* saturate near 1 for all reasonable charts, flag only gross violations (density, strain, overload, spike), and act as vetoes—they never average into a score; *diagnostics* carry signal with direction-only interpretation (the fixed-lattice phase witness; the hypersensitive rare- n -gram rate). We report score profiles, not a single aggregate: fixing trade-off weights between axes is exactly the judgment we defer to human data.

Family	measures	inputs	type	tier
timing	placement vs. grid	times + grid	defect	diag
grammar	sequence idiom	tokens + LM	both	rank
structure	learnable organization	tokens	typicality	rank
coupling	chart–music dialogue	times + audio	typicality	rank
gap	distance to human manifold	features	typicality	rank
constraints	playability limits	tokens	defect	vetc

Table 1. The five-plus-one metric families. *Inputs* determine cross-vocabulary applicability (Section 5.5): timing and coupling read note times only and apply to any rhythm game; grammar, structure, and gap require a vocabulary tokenizer and a calibration corpus. Perceptual constants are fixed from the temporal-discrimination literature [12, 13]: a 6 ms absolute deadzone equal to the relative-distortion floor, a 2.5% Weber fraction of the local inter-onset interval, severity tiers at $1 \times / 2 \times / 3 \times$ the deadzone. LM: trigram, $\alpha=0.05$; bands: per-course $[p_{10}, p_{90}]$.

Probe (dose knob)	Damages	Designated target (disclosed)
C1 timing jitter ($\pm 10/20/30$ ms)	micro-timing	rhythm complexity
C1s sparse outliers (.5/1/2%, ± 60 ms)	timing tail	placement
C2 anchor shift (+15/30/60 ms)	audio alignment	timing witness
C3 type shuffle ($p=.2/.4/.8$)	note grammar	transition validity
C4 loop collapse ($p=.3/.6/1$)	structure	repetition adequacy
C5 blandification ($p=.3/.6/1$)	interestingness	(anti-gaming) exhibit
C6 density scale ($\times .5/1.5/2$)	difficulty fit	density adequacy
C7 burst insert (2/4/8 clusters)	playability	density-spike score
C8 bar shuffle ($p=.3/.6/1$)	global form	repetition adequacy

Table 2. Nine corruption probes applied to 170 official charts at three doses each (4,760 scored variants incl. originals; deterministic seeds; operators verified by independent re-execution). Every probe preserves what it does not target: C3–C5 preserve all note times exactly; C4 and C6 preserve or control density. C1s displaces $\max(1, \lceil np \rceil)$ notes by ± 60 ms ($10 \times$ the deadzone), the “three ruined notes in a thousand” failure mode that mean statistics cannot see.

4. THE CORRUPTION-PROBE HARNESS

Table 2 lists the probes. Corruption direction is guaranteed at the population level: since band scores are two-sided, a corruption can push an individual off-band chart *toward* its band, so all criteria are evaluated over 170 charts, with per-chart paired direction (the fraction of charts moving against their own intact version) as the certification statistic—magnitude medians can saturate under re-matching effects, paired direction does not. Acceptance requires (a) *sensitivity*: the designated target degrades with dose; (b) *anti-gaming*: no score family rewards damage without bound; and (c) *disclosed coupling*: the full probe $\times$ metric response matrix is published rather than summarized by its diagonal.

Fairness rules fixed in advance: identical probes for baselines and calibrated scores; identity pairs across layers (distinct-4 $\equiv$ unique-4-gram rate; transition validity $\leftrightarrow$ unseen-transition rate) are never counted as independent agreement; length confounds are checked against note count; and the blandification attack is additionally run

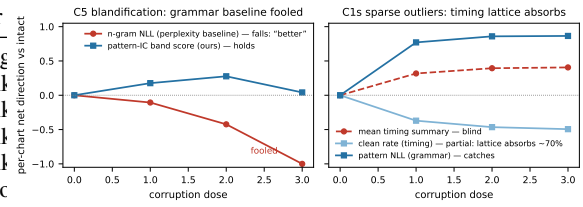


Figure 1. Bidirectional blindness, as per-chart paired net direction (fraction of charts moving up minus moving down vs. their own intact version). **Left (C5 blandification):** rewriting notes toward the LM argmax drives n -gram NLL *down* in essentially every chart (net direction -1.0 at full dose)—“better” under the standard lower-is-better reading—while the band score built on the same LM statistic stays near zero net direction: it saturates at typicality instead of rewarding the damage. **Right (C1s sparse outliers):** displacing 0.5–2% of notes by ± 60 ms leaves mean timing summaries at exactly zero median response (disclosed), moves the timing clean rate in only a fraction of charts (the subdivision lattice re-matches $\approx 70\%$ of displaced notes within the deadzone), but flips the IOI token class of adjacent intervals, so grammar NLL responds with net direction $+0.86$ even at one to three displaced notes per chart. Timing damage that placement metrics absorb is caught as grammar damage.

cross-LM: the attack is driven by an LM trained on half the calibration songs and measured by an LM trained on all of them, so perplexity’s failure cannot be an artifact of attacking its own model.

5. RESULTS

5.1 Sensitivity: targeted damage moves the right scores

All six probes with calibrated band targets pass: dose–response Spearman $\rho(\text{dose}, \text{target score})$ is negative for C1 rhythm complexity (-0.48), C3 transition validity (-0.33), C4 repetition adequacy (-0.41), C6 density adequacy (-0.68), C7 density spike (-0.79), and C8 repetition adequacy (-0.48), with monotone mean trajectories (all $p < 10^{-18}$). C2’s target is the timing witness (Section 5.3); C1s’s response pattern is the complementarity exhibit (Section 5.4); C5’s bounded response is the anti-gaming property, reported next.

5.2 Reductio: borrowed metrics reward damage

Figure 1 (left) and two corpus numbers give the reductio exhibits. *Perplexity rewards blandness*: official test charts average perplexity 9.4 under the corpus trigram; flattening them toward the LM’s modal patterns drives perplexity down 37% ($9.44 \rightarrow 5.98$). Read as “lower is better” [1], perplexity prefers the damaged chart—a corpus-LM instance of the likelihood/quality dissociation [10]—and the reversal survives the cross-LM version of the attack. The band score over the same statistic cannot be driven above

typicality (bounded at 1), and the damage the attack actually does is caught where it lands: repetition adequacy falls $0.99 \rightarrow 0.84$. *Structurelessness rewards collapse*: replacing charts with their own dominant 4-gram loop raises self-similarity structureness [9] by 62% ($0.085 \rightarrow 0.137$)—copy-paste is maximally self-similar—and set-level Self-BLEU rises in lockstep ($0.845 \rightarrow 0.898$), while repetition and variety adequacy correctly fall ($0.99 \rightarrow 0.80$, $0.99 \rightarrow 0.89$).

Two observational results complete the case. *Direction instability*: against the two published taiko generators of Section 5.5, whose outputs are noisier than the human corpus, perplexity separates official from generated charts in the correct direction (AUC 0.80)—yet C5 proves the same reading inverts for outputs blander than the corpus. A signed quality metric whose sign depends on which side of the human distribution the system happens to sit is not usable as a ranking signal; band scores are stable on both sides by construction. *Length confounds*: on the same pool, type-token ratio correlates with note count at $|\rho| = 0.85$ and distinct-2 at 0.56, against ≤ 0.28 for every calibrated ranker.

5.3 A perceptually anchored timing family

Timing is the one family that cannot be measured from the chart alone: it needs a grid reference. We fix a two-tier hierarchy: (1) the *authored* grid—per-bar timing metadata (bar lines, per-bar meter) shipped with the song’s official charts, which carries no note positions and is therefore fair for reference-free evaluation; (2) an *estimated* grid from an audio beat tracker, for songs without authored metadata. A generated chart’s self-declared grid is never consulted: a system that supplied its own reference could trivially zero its own offsets. On our corpus the estimated tier costs little: estimated beat periods agree with authored tempo within 2% on 40/40 test songs, and phase-sensitive statistics remain usable through the witness below.

All thresholds are perceptual, not game-specific: a single 6 ms deadzone—the just-noticeable timing distortion for short intervals [12]—below which deviations are treated as imperceptible; severity tiers at $1 \times /2 \times /3 \times$ the deadzone; and a relative channel that compares consecutive inter-onset intervals against a Weber envelope $\max(6 \text{ ms}, 2.5\% \times \text{IOI})$, computed on raw offsets before any deadzone is applied. The family reports a partition of all notes—*clean* (matched within the deadzone), tier exceedances, and unmatched notes—so that the primary statistic, the *clean rate*, has unmatched notes in its denominator and cannot be improved by losing notes. Aggregation is per-chart first, never pooled over notes: a chart is ruined by its own worst notes, and cross-chart tails are reported alongside medians.

Under dense jitter (C1) the family certifies cleanly: the clean rate falls in 100% of charts at every dose, and the first-tier exceedance at $\pm 10 \text{ ms}$ jitter matches its theoretical value ($P(|U(-10, 10)| > 6) = 0.4$; measured median 0.39).

Closing the anchor blind spot. A chart can be id-

iomatic on every chart-only axis and still be shifted off the music entirely—C2 makes this concrete. A constant $+15/30/60 \text{ ms}$ shift of every note changes all 56 chart-only quantities by at most 0.03 SD, and the obvious fix is equally blind: a matching-based signed offset (match each note to its nearest lattice anchor, average the residuals) re-matches every shifted note on a dense subdivision lattice. Detecting the shift requires a *fixed*-lattice witness: the global offset that maximizes support of the note stream on a lattice laid inside each authored bar (eighth-note, beat, and bar-line scales combined), searched over $\pm 250 \text{ ms}$ with minimum-norm tie-breaking. Because the lattice follows the authored bar lines it is aperiodic under tempo changes, and the witness recovers the injected shifts with exact medians (15.00/30.00/60.00 ms) on $\geq 99.4\%$ of charts on *both* grid tiers. The same witness applied to published systems is what detects their anchoring behavior in Section 5.5.

5.4 Complementarity: the profile is not compressible

Three probes, three different blind/catch splits (Figure 1, Figure 2):

- **C2 anchor shift**: invisible to all 56 chart-only quantities and to matching-based offsets; caught only by the timing witness.
- **C1s sparse outliers**: largely absorbed by lattice matching ($\approx 70\%$ of displaced notes re-match within the deadzone; mean summaries show exactly zero median response); caught by the grammar family (net direction $+0.86$ to $+0.88$), because a 60 ms displacement necessarily flips the IOI token class of two adjacent intervals. Density and structure stay silent ($|\text{net direction}| \leq 0.02$)—the response is specific, just not where the probe was aimed.
- **C5 blandification**: fools the grammar baseline (perplexity), is bounded by the band layer, and is caught by the structure family.

Each family has a corruption it cannot see and a corruption only it sees. Any scalar aggregate would have to fix exchange rates between axes that these results show are not exchangeable.

5.5 Application: five published systems, three vocabularies

We evaluate five published generators spanning three chart vocabularies: two taiko-domain systems—Mapperatorinator [14], an open-source osu!-ecosystem generator run in taiko mode [PENDING: E: positioning wording], and TaikoNation [2]—plus, for the game-agnostic families, Dance Dance Convolution’s onset pipeline [1], the public timing-only configuration of GenéLive! [3], and an osu! difficulty-targeted generator [15]. All systems chart the same 40 held-out songs from audio; none was trained on this corpus. Figure 3 and

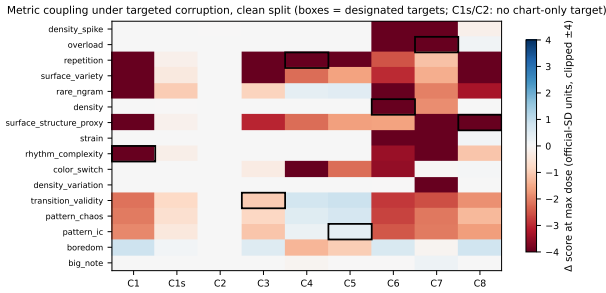


Figure 2. The full probe $\times$ score response matrix on the clean test set (official-SD units at max dose, clipped at ± 4 ; boxes = designated targets; C2 and C1s have no chart-only target). Targeted probes move their targets, but responses are *not* diagonal: timing jitter (C1) propagates into pattern scores because pattern tokens are IOI-quantized—in a rhythm game, timing damage *is* pattern damage. We publish the matrix as a coupling map rather than claiming orthogonality; band-layer cells for C1s are pale because one-to-three-note damage is sub-band, and its detection lives in the raw diagnostic layer (Figure 1), which is the division of labor the role taxonomy prescribes.

Figure 4 give the profiles; three findings illustrate what they add over single numbers.

Same score, different diseases. The two taiko systems have statistically indistinguishable timing clean rates (0.59 vs. 0.58), yet opposite mechanisms. Mapperatorinator is internally coherent (relative-channel violations 0.10, close to officials) but phase-misanchored: the witness reads a 26 ms median offset, while the matching-based statistic reads under 1 ms—the C2 lesson live on a published system. TaikoNation is incoherent on both channels (relative violations 0.41; witness 55 ms) and, uniquely among the five, deaf to the music: its density–energy correlation is -0.003 and its notes land on average-energy frames (at-hit envelope $z = -0.004$, vs. 1.09 for officials). The onset pipeline anchors accurately (2.5 ms), which doubles as a control: all five systems charted the same audio files, so the 26–55 ms offsets are model behavior, not pipeline artifacts.

Complementary failures, invisible to any scalar. On chart-vocabulary families the two taiko systems are near-perfect complements (Figure 4). Mapperatorinator writes in-band pattern language (repeated-4-gram rate 0.51 against a human band of 0.45–0.72; repetition and variety adequacy 1.0) but fails difficulty conditioning: its oni and ura courses sit inside their density bands while its easy/normal/hard courses run 1.5–2.5 $\times$ over theirs (easy: 2.9 notes/s against a band of 0.8–1.8)—its “easy” is not a taiko easy, a failure only per-difficulty calibration can see, and one that trips the playability gate. TaikoNation is the mirror image: density and strain in band, but pattern language collapsed (perplexity 161 vs. 9.4 for officials; 55% of transitions never occur in the human corpus; repeated-4-gram rate 0.004—no learnable repetition at all).

Progress statements need profiles. Between TaikoNa-

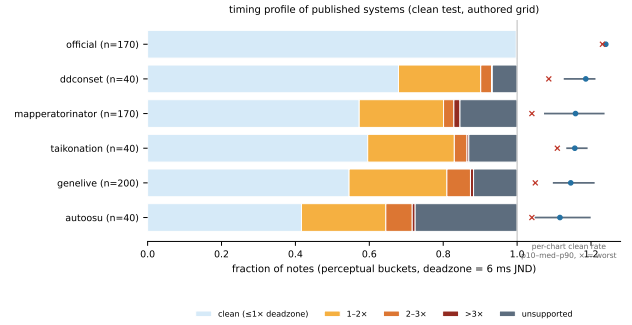


Figure 3. Timing profiles of published systems on the clean test set (authored grid). Each bar partitions all notes of a system into perceptual buckets: clean ($\leq 1\times$ the 6 ms deadzone), tier exceedances, and unmatched. Right strip: per-chart clean-rate p10–median–p90 with the worst chart marked $\times$ —tail charts that corpus means hide. Official charts are the all-clean reference row.

	density	energy	strain	repetition	variety	pattern	onset	clean	1-2x	2-3x	>3x	unsupported
official	1.00	1.00	1.00	0.75	1.00	1.00	1.00	✓	✓	✓	✓	✓
mapperatorinator	0.59	0.54	0.37	0.23	0.01	1.00	1.00	✓	✓	✓	✓	✓
taikonation	0.59	1.00	1.00	0.00	0.00	0.00	0.00	✓	✓	✓	✓	✓
ddconset	0.71	—	—	—	—	—	—	—	—	—	—	—
geneline	0.53	—	—	—	—	—	—	—	—	—	—	—
autoosu	0.58	—	—	—	—	—	—	—	—	—	—	—

Figure 4. System profiles and applicability matrix. Cells are 0–1 scores (timing clean rate; calibrated band scores); grey dashes mark families that do not apply to the system’s chart vocabulary; glyphs are constraint gates (pass/fail; a failed gate outlines the row: gates veto, they do not average in). The two taiko-vocabulary systems fail in complementary ways—one row’s dark cells are the other’s pale cells—which is the profile argument on real systems.

tion (2021) and Mapperatorinator (2024) the timing clean rate is unchanged—a single-number comparison would report three years of no progress. The profile shows where the progress went: sequence coherence improved by a factor of four (relative violations 0.41 $\rightarrow$ 0.10) and pattern language moved from collapsed to in-band, while anchoring and difficulty calibration remain open problems.

6. DISCUSSION

Out-of-band is a flag, not a verdict. The probes license a one-directional inference: damage drives charts out of band. The converse is not certified, and should not be—a generator that produces charts outside the human envelope while passing the constraint gates, coupling to the music, and retaining learnable structure is a candidate for novelty, not a failure. In the creativity framing of novelty-plus-value, the typicality axes measure novelty’s magnitude while the defect-type axes and the rest of the profile supply the value judgment; the same manifold-distance reading means “damage” when the value axes collapse and “originality to review” when they hold. This adjudication is itself an argument for profiles: the meaning of one axis

depends on the others.

Beyond one game. Two families read only note times and were applied unchanged across three chart vocabularies here (timing, coupling); the token families (grammar, structure, gap) transfer with a vocabulary adapter—lane, position-bin, or direction tokens replace color tokens—plus a calibration corpus of human charts for the target game, and the constraint family re-calibrates its limits from the same corpus. The probes themselves operate on (t, class) event streams and port as-is, so the certification protocol, not any single metric, is the transferable artifact: given a new game’s corpus, the same harness certifies which metrics work there. Position-based games (4K, osu!, maimai) additionally need a spatial-ergonomics family (movement cost, hand allocation, rotation) that no chart-vocabulary adaptation of the present families supplies; we leave it as the main structural extension.

7. LIMITATIONS AND FUTURE WORK

No human validation. The probes certify damage detection, not experience prediction; validating the surviving metrics against player judgments—including the novelty-vs-damage adjudication of Section 6—remains future work. **Population-level direction.** A minority of individual charts move toward their band under corruption; per-chart verdicts need the profile, not one score. **Coupling.** The suite is a coupled profile (Figure 2); attribution between timing and pattern damage is smeared by shared tokenization, which is partly domain truth (mistimed notes damage patterns as played). **Scope.** One game corpus calibrates the bands; the cross-game systems are evaluated on the game-agnostic families only; systems are run in their public inference configurations at one seed.

8. CONCLUSION

Chart generation needs evaluation it can trust more than it needs another metric. Corruption probes make metric validation systematic and reproducible: the borrowed metrics did not merely underperform—they rewarded blandness, rewarded collapse, missed a fatal global misalignment entirely, and flipped ranking direction across system regimes. Corpus-calibrated band scores over the same raw quantities, extended with a perceptually anchored timing family over a grid reference hierarchy, pass the same probes—population-exact dose response, bounded response to the gaming attack, exact shift recovery—and their coupling structure is published rather than hidden. Applied to five published systems, the resulting profiles replace rankings with mechanism diagnoses: same-score systems with opposite diseases, progress that single numbers cannot see, and difficulty-conditioning failures only per-difficulty calibration detects. The harness, calibration artifacts, per-chart records, and evaluation package are released so that future chart-quality metrics—for this game or others—can be validated under the same protocol. **[PENDING: P8: artifact URL.]**

9. ETHICS STATEMENT

The corpus consists of community-created charts used under research terms; no personal data is involved. Better automated evaluation chiefly benefits players and small developers; we see no direct misuse vector beyond automated content farming, which platform moderation already addresses.

10. AI USAGE STATEMENT

AI assistants were used for experiment orchestration, drafting, and editing under author direction; all numbers were produced by the released code, adversarially re-derived, and verified by the authors.

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